INCH-POUND

MIL-F-70546 (AR)
AMENDMENT 1
28 September 1994

MILITARY SPECIFICATION

FIN ASSEMBLIES, MORTAR CARTRIDGE, 81MM, M28, M29

This Amendment forms a part of MIL-F-70546 (AR), dated 5 September 1986, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.2, following E-8 Tension Testing of Metallic Materials:

Add: "E-340 Macroetching Metals and Alloys".

PAGE 3

2.3, Third line:

Delete: "detain", and substitute: "detail".

3.4, and wherever it appears in the specification:

Delete: "housing" or "fin housing",

and substitute: "cartridge housing".

PAGE 5

4.3.1, First Article, Part Description: Add the following:

"Tensile specimen from cartridge housing stock	See ASTM E-8	2
Tensile specimen from fin stock	See ASTM E-8	2
Microstructure specimen from cartridge housing tensile specimen	----	2
Microstructure specimen from fin stock tensile specimen	----	2
Macroetch specimen from fin stock	----	1"

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PAGE 6

Under Fin, after Tension Test, add the following:

"Macrostructure examination	1 1/	100%	3.3	4.5.6/Visual
Microstructure examination	2	100%	3.3	4.5.7/Visual"

Under Housing, Cartridge, after Tension Test, add the following:

"Microstructure examination	2	100%	3.3	4.5.7/Visual"

Under Note, add: "1/ Examination performed on bar stock".

PAGE 8

4.4.2.1, Major 101, under Sample Units, add: "1/".

Under AQL, delete: "100%".

Add: Major 102, as follows:

"102	Microstructure examination	2/	(-)	3.3	4.5.7/Visual"

Under Note, add:

"1/ See 4.5.5.1. All specimens must pass.
2/ One per tension test specimen".

PAGE 11

4.4.2.2, Major 130: Delete in its entirety.

PAGE 12

4.4.2.3, after 9327834, in Drawing Number box, add "or 9381092".

Major 106, delete: "rear chamber",

and substitute: "start of rear OD".

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PAGE 13

4.4.2.3, Major 119, under Sample Units, add: "1/".

Under AQL, delete: "100%".

Add: Major 120, as follows:

"120 Microstructure examination 2/ (-) 3.3 4.5.7/Visual"

Under Note, add:

"1/ See 4.5.5.2. All specimens must pass.
2/ One per tension test specimen".

PAGE 16

4.5.5: After 4.5.5 Tension test., insert the following:

"The materials shall be certified from the aluminum vendor as meeting the applicable mechanical properties. In addition, the contractor shall have the following samples tested independently from the aluminum vendor. Samples shall be selected by the contractor after receipt of the certified material."

4.5.5.1: Wherever it appears, delete: "R", and substitute: "round".

Add new paragraphs 4.5.6 and 4.5.7 as follows:

"4.5.6 Macroetch test. Following nominal cropping full cross section macroetch specimens shall be obtained from the rear of each extruded length, regardless of extrusion method. Additionally, the front of the first acceptable extruded stock from the billet which represents the top of each ingot ("log") shall be macroetched as above. If the front/rear identity of each billet is not maintained, or if the top billet from an ingot is not distinguished from successive billets from the ingot, then macroetch specimens shall be obtained from both ends of each billet. Defective material shall be discarded until the requirements of 3.1 are met. Macroetching shall be in accordance with ASTM Method E-340, except sodium hydroxide shall be the reagent used."

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"4.5.7 Microstructure test. From each of the samples submitted for tension testing, one wafer, approximately one-half inch square, shall be prepared for metallographic examination from an undeformed portion of the specimen. The specimen shall be examined in the direction which is perpendicular to the extrusion flow. When suitably prepared, the specimen shall be etched with Keller's reagent and examined by scanning under a metallurgical microscope at a minimum magnification of 500 diameters. Overheating shall be evidenced by the presence of rosettes in the microstructure or incipient melting at the grain boundaries. Evidence of the above shall cause rejection of all material represented by the applicable sample."

6.1, Wherever it appears, delete: "M853", and substitute "M853A1".

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